

Linear Algebra 2

Vincent Wang

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1 Introduction

1. Let $f_1, f_2, \dots, f_p : E \rightarrow E$ be linear maps on $E = K^n$ with $\text{char } K \neq 2$ such that $\sum f_i = \text{id}_V$, $f_i \circ f_j \equiv 0$ for $i \neq j$. Then f_i are idempotent.

Let $M_i \in M_n(K)$ represent each f_i . Note

$$I = \left(\sum M_i \right)^2 = \sum M_i^2 + \sum_{i \neq j} M_i M_j = \sum M_i^2.$$

Further,

$$\begin{aligned} M_i^2 &= \left(I - \sum_{j \neq i} M_j \right)^2 = I - 2 \sum_{j \neq i} M_j + \left(\sum_{j \neq i} M_j \right)^2 \\ &= I - 2 \sum_{j \neq i} M_j + \sum_{j \neq i} M_j^2 = 2 \left(I - \sum_{j \neq i} M_j \right) - M_i^2 \\ &= 2M_i - M_i^2, \end{aligned}$$

so $2M_i = 2M_i^2 \rightarrow M_i = M_i^2$, so f_i is idempotent.

2. Let f_i be as before. Let $U_i = f_i(E)$. Then the U_i are independent.

Suppose $f_1(v_1) + f_2(v_2) + \dots + f_p(v_p) = 0$. Then $f_k \circ (f_1(v_1) + f_2(v_2) + \dots + f_p(v_p)) = f_k(0) = 0 \rightarrow f_k \circ f_1(v_1) + f_k \circ f_2(v_2) + \dots + f_k \circ f_p(v_p) = 0 \rightarrow f_k(v_k) = 0$.

3. Suppose V/K has infinite dimension. Let $V = K^J$. Each map $v^* \in V^*$ does not necessarily correspond with a vector $v \in V$. Ex: the sum of the components is given by $v^* = (1)_{\alpha \in J}$, but $v \notin V$.

4. For finite dimension, $V \cong V^*$.

5. Suppose $U < V$. Define $U^\circ = \{v^* \in V^* : v^*(U) = 0\}$. For finite dimensional V , $U^\circ \cong U^\perp$, so $\dim U^\circ + \dim U = \dim V$.

6. Let K be a field and J a set. Denote by K^J the set of all tuples $v = (v_\alpha)$, where $v_\alpha \neq 0$ for finitely many α . For $v \in K^J$ denote by $J_v = \{\alpha \in J : v_\alpha \neq 0\}$ a finite set. Let addition and scalar multiplication be component-wise.

(a). K^J is closed under addition and scalar multiplication. Suppose for $v, w \in K^J$ that $(v + w)_\alpha = v_\alpha + w_\alpha \neq 0$. Then $v_\alpha \neq 0$ or $w_\alpha \neq 0$, so $\alpha \in J_v \cup J_w$, which is finite. If for $k \in K$, $(kv)_\alpha = kv_\alpha \neq 0$, then $v_\alpha \neq 0$, so $\alpha \in J_v$.

(b). $(K^J, +)$ is an Abelian group. Note $\vec{0} \in K^J$, so $K^J \neq \emptyset$. Also, for $v, w \in K^J$, $(v + w)_\alpha = v_\alpha + w_\alpha = w_\alpha + v_\alpha = (w + v)_\alpha \rightarrow v + w = w + v$. $(\vec{0} + v)_\alpha = 0 + v_\alpha = v_\alpha \rightarrow \vec{0} + v = v$. Let $(-v)_\alpha = -v_\alpha$. Note the $-$ operation is closed in K^J . Note $(v + (-v))_\alpha = v_\alpha + (-v)_\alpha = v_\alpha + (-v_\alpha) = 0 = \vec{0}_\alpha \rightarrow v + (-v) = \vec{0}$. If $u, v, w \in K^J$, $((u + v) + w)_\alpha = (u + v)_\alpha + w_\alpha = (u_\alpha + v_\alpha) + w_\alpha = u_\alpha + (v_\alpha + w_\alpha) = u_\alpha + (v + w)_\alpha = (u + (v + w))_\alpha \rightarrow (u + v) + w = u + (v + w)$.

(c). K^J is a K -module. Show the distributivity laws.

(d). $e_\alpha \in K^J$ where $(e_\alpha)_\alpha = 1$ and 0 otherwise is a basis for K^J (by construction).

(e). Let $V = \mathbb{R}^\omega$ have the product space. Then $V^* = \mathbb{R}^\omega$ has the box space. I don't think V^* has a countable basis, so $V^* \not\cong V$.

Let K be a field. K^ω has an uncountable basis, so $K^\omega \not\cong K^\infty$.

Suppose FTSOC K^ω has a countable basis $\{x_1, x_2, x_3, \dots\}$. For $x \in K^\omega$, define $\pi_{[m, n]} : K^\omega \rightarrow K^{n-m+1}$ by selecting the coordinates of x from indices m to n , inclusive. Let $l \in \mathbb{N}$ be given. Let $m_l = l(l+1)/2$ be the l th triangular number and $n_l = (l+1)(l+2)/2 - 1$ one less than the $l+1$ th triangular number, so for every natural there exists a unique l for which $m_l \leq a \leq n_l$. Also, $m_l - n_l + 1 = l + 1$.

Let $B_l = \pi_{[m_l, n_l]}(\{x_1, x_2, \dots, x_l\}) \subset K^{l+1}$. Since there are l x 's, $\dim \text{span}(B_l) \leq l$, so we may pick $y_{[m_l, n_l]} \in K^{l+1} - \text{span}(B_l)$. Since the intervals $[m_l, n_l]$ partition $\mathbb{N}$, y is uniquely determined (once the $y_{[m_l, n_l]}$ are determined).

Suppose FTSOC $y = \sum a_i x_i$ for $i \in M$ a finite subset of $\mathbb{N}$. Let $l \in M$ be maximal. Then $y \in \text{span}(x_1, x_2, \dots, x_l)$, so $y_{[m_l, n_l]} \in \pi_{[m_l, n_l]}(\text{span}(x_1, x_2, \dots, x_l)) = \text{span } B_l$, contradiction.

Let V/K be finite dimensional, $\text{char } K \neq 2$, $\alpha : V \times V \rightarrow K$ a symmetric bilinear form. There exists an orthogonal basis of V wrt α .

Let $B_0 = \emptyset$, $V_k = \text{span } B_k$. Suppose for induction (with $n < k$) B_k is or-

thogonal independent, $\alpha(e_i, e_i) \neq 0$ for all $i \leq k$ ($e_i \in B_k$), and $V = V_k \oplus V_k^\perp$.

Suppose $\alpha|_{V_k^\perp} \equiv 0$. Then complete B_k to a basis $\{e_1, e_2, \dots, e_n\}$ by choosing $e_i \in V_k^\perp$ for $i > k$ as a basis of $V_k^\perp$. Else, $\alpha \neq 0$, so because $\text{char } K \neq 2$, pick $e_{k+1} \in V_k^\perp$ such that $\alpha(e_{k+1}, e_{k+1}) \neq 0$. Note B_{k+1} is independent and orthogonal.

It remains to show $V_{k+1} \cap V_{k+1}^\perp = 0$. Let $x = \sum x_i e_i \in V_{k+1}$ such that $\alpha(x, e_i) = 0$ for all $i \in [k+1]$. Then $x_i \alpha(e_i, e_i) = 0$ for all $i \in [k+1]$, so $x_i = 0$ for all i , so $x = 0$.